Xi'an Jiaotong-Liverpool University

西交利物浦大学

CPT111 PAST FINAL EXAMINATION

BACHELOR DEGREE – Year 2

INTRODUCTION TO PROGRAMMING IN JAVA

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

1、 **The final exam will count for 40% of the final module mark.**

2、 **Answer all questions.**

3、 **All questions should be answered on the Learning Mall Quiz Page.**

4、 **All the answers must be in English.**

CPT111 Past Final Exam

Part 1 Multiple Choice

```
String type = "reptile";
int legs = 6;
boolean dangerous = true;
String pattern = "striped";

    if (type.equals("mammal")) {
      if (legs == 4) {
        if (dangerous == true) {
          if (pattern.equals("solid")) {
            lion();
          } else {
            tiger();
          }
        } else {
          cat();
        }
      } else {
        System.out.println("Not classified");
      }
    } else if (type.equals("reptile")) {
      if (legs == 4) {
        if (dangerous == true) {
          crocodile();
        } else {
          lizard();
        }
      } else if (legs == 0) {
        snake();
      } else {
        System.out.println("Not Classified");
      }

    } else if (type.equals("insect")) {
      if (legs == 8) {
        spider();
      } else if (legs == 6) {
        beetle();
      } else {
        System.out.println("Not Classified");
      }
    } else {
      System.out.println("Not Classified");
    }
```

1. When the code above is run, what method is called?

   a) The tiger() method is called.
   b) The "cat()" method is called.
   c) No method is called, but "Not Classified" is displayed.
   d) The beetle() method is called.

2. Based on the code above, what is the correct description for the if statement given below?

```
if (type.equals("reptile")) {
   if (legs == 4) {
      if (dangerous == true) {
         crocodile();

} } }
```

   a) If the type is equal to the value of "reptile", and if the number of legs is equal to four, and if the boolean variable dangerous is set to false, then call the crocodile method.

   b) If the type is equal to the value of "reptile", and if the number of legs is equal to six, and if the boolean variable dangerous is set to false, and if the pattern variable is set to "solid", then do not call the crocodile method.

   c) If the type is equal to the value of "reptile", and if the number of legs is equal to four, and if the boolean variable dangerous is not set, then call the crocodile method.

   d) If the type is equal to the value of "reptile", and if the number of legs is equal to four, and if the boolean variable dangerous is set to true, then call the crocodile method.

3. Another way to write the if statements above is to use multiple condition if statements. If we keep the condition values the same, what code will correctly call the cat() method?

   a) if (type.equals("mammal") && legs == 4 && dangerous == false) {
   b) if (type.equals("mammal") && legs == 4 && dangerous) {
   c) if (type.equals("mammal") && legs == 4) {
   d) if (type == "mammal" && legs == 4 && dangerous==true) {

4. How many times will this for loop run?
   a) 1 time
   b) Will not run
   c) 5 times

```
for(int i = 5; i > 1; i --){

  System.out.println("Value is " + i);
  counterDisplay(i);

}
```

d) 4 times

5. This while loop will run 0 times. Why?

```
double total = 100;
while (total<5){
   total = total -10;
 }
System.out.println(total);
```

a) The double variable is not declared.
b) The condition is never true.
c) The condition should be <= 5.
d) The use of curly brackets is not correct.

6. Given a double variable, double input = 3.1;, how would you cast the double variable to an integer?

a) (double) input;
b) (int) input;
c) Integer.toString(input);
d) Double.parseDouble(input);

7. Using the calculations on the right, after they have been completed, what will the final value of **x** be?

```
int x = 3
int y = 4
int z = 4

x = x*y
y = x/z
z = x + z/y
x = z%y + y
```

a) 4
b) 7
c) 4.5
d) 1

8. Using the calculations on the right, after they have been completed, what will the final value of **y** be?
a) 0
b) 4
c) 0.25
d) 3

9. Using the calculations on the right, after they have been completed, what will the final value of **z** be?
a) 5
b) 4
c) 14
d) 13

10. What Java principle is being demonstrated by the following line of code?

    public class Hotel extends Travel {

    a) Encapsulation
    b) Method Overloading
    c) Inheritance
    d) Constructors

```
public class Hotel extends Travel {

    private double cost;

    public Hotel(double cost) {
        super() ;
    }

    public int getCost(){}

    public boolean equals(Object h) {}

    public String toString() {}
}
```

11. What type of variable is the following code?

    private double cost;

    a) Static variable
    b) Local variable
    c) Public variable
    d) Instance variable

12. What is the name of the superclass in the code example above?

    a) Hotel
    b) Travel
    c) Constructor
    d) Class

13. What does the following line of code do?

    super();

    a) A new Hotel object is created
    b) The constructor of the Hotel object is called
    c) The constructor of the Travel object is called
    d) The variables are reset

14. What Java principle is being demonstrated by the following line of code?

    public boolean equals(Objectl h) {}
    public String toString() {}

    a) Instance variables
    b) Method Overloading
    c) Method Overriding
    d) Parameter Passing

15. What Java principle does the following code demonstrate:

```
private double cost;
public int getCost(){}
```

a) Encapsulation
b) Shared variables
c) Inheritance
d) Method overloading

16. Which of these statements is correct:

a) A Java file is interpreted on the Java Virtual Machine.
b) A Java file is compiled into bytecode, and then the Java Virtual Machine compiles and runs the bytecode file.
c) A Java file is compiled into bytecode, and then the Java Virtual Machine interprets the bytecode file.
d) A Java file is compiled into assembly language, and then the Java Virtual Machine interprets the assembly file.

17. Which is a valid declaration and initialization of a float?

a) float f = 2.1f;
b) float f = "2.1f";
c) float f = new 2.1f;
d) float f = 2.1;

18. Given a circle of radius 3, what is the correct Java code to calculate the area of a circle?

a) double area = Math.pow((Math.PI *3), 2);
b) double area = Math.PI *Math.pow(2, 3);
c) double area = Math.PI *Math.pow(3, 2);
d) double area = 2 * Math.PI *3;

19. Given the String input below, how would we produce the output "Exam"?

```
String input = "CPT105 Exam Today";
```

a) input.substring(7, 11);
b) input.substring(0,6);
c) input.substring(6, 11);
d) input.substring(7);

20. Given the String input below, how would we produce the output "TODAY"?

String input = "CPT105 Exam Today";

a) input = input.toUpperCase (input.length()-5, input.length()).equalsIgnoreCase();
b) input = input.substring(input.length()-5, input.length()).toUpperCase();
c) input = input.toUpperCase (input.length()-5, input.length());
d) input = input.substring(7, 11).toUpperCase();

21. What Java Class would you use to calculate the difference between dates?

a) The Date Class
b) The Period Class
c) The Duration Class
d) The getDate() Class

22. Which line of code would correctly instantiate a new LocalDateTime object?

a) LocalDateTime dateTime = LocalDateTime.of("2020, 11, 5, 9, 30");

b) LocalDateTime dateTime = LocalDateTime.of(2020, 11, 5, 9, 30);

c) LocalDateTime dateTime = LocalDateTime.of(2020, 11,5);

d) LocalTime getTime = LocalDate.getTime(22, 40, 10);

23. We wish to create a method named calculate that will receive an integer and a double as arguments and return an integer. Which method header is correct?

a) public static int calculate(int a, int b);
b) public static void calculate(int a, double b){
c) public static double calculate(double a,b){
d) public static int calculate(int a, double b) {

24. We wish to create a method named checkAndDisplay that will receive 2 boolean arguments, but not return anything. Which method header is correct?

a) public static void checkAndDisplay (boolean a, boolean b){
b) public static void checkAndDisplay (boolean a, boolean b) ;
c) public static boolean checkAndDisplay (boolean a) {
d) public static boolean checkAndDisplay(String a, String b){

25. Looking at the code on the right, what is represented by B?

```
        [B]
private int getFullNum (int idNumber){
}
```

a) Access modifier for the method
b) Argument or parameter to the method

c) Return type for the method
d) Return value for the method

Part 2 Code Completion

Question 1 (10 marks)

Look at the code below, read it and then complete the code description below by filling in the spaces.

```
public static String viewVal() {
    String[] arr = {"First Val", "Second Val", "Third Val"};

    Scanner sc = new Scanner(System.in);
    System.out.println("Choose Index to view");

    int choice = 0;

    try {
      String in = sc.nextLine();
      choice = Integer.parseInt(in);

    } catch (NumberFormatException e) {
      System.out.println("Non integer entered");
      return "Try again";
    }

    String chosenString = "Not chosen yet";
    try {
      chosenString = arr[choice];
      System.out.println("You chose " + chosenString);
    } catch (ArrayIndexOutOfBoundsException e) {
      System.out.println("Chose an index outside array bounds");
      return "Try again";
    }
    return chosenString;
  }
```

This is a method named **viewVal** that allows the user to choose a value in an array using keyboard input. It has no parameters and returns a **String**. It is not part of an object and is a **static** method. The method first creates an array of Strings of length 3, and assigns values. We then declare and initialise a **Scanner** object, and display a message to the user to instruct them to choose a suitable index. The input will be **wrapped** in a try catch, and the String will be **parsed** to an integer. If the String does not parse, a **NumberFormat** Exception is thrown, which will then be caught and an error message will be displayed to the user.

If the input is correctly parsed to an integer, then the method attempts to select the appropriate value from the array. It will attempt to assign and display the string using the index entered by the user. If the user inputs a number larger than the size of the array, an **ArrayIndexOutOfBounds** Exception will be thrown and **caught**, with an error message displayed to the user.

Finally, if a valid input has been received, the appropriate String will be **returned** from the method.

Question 2 (20 marks)

a) Create a class to store data about a book. A book has a title, an author, and a number of pages. You should use sensible variable types and names, and create instance variables, a constructor, appropriate encapsulation, and override the inherited methods (10)

```
public class Book {
}
```

2 marks – private instance variables, 2 strings and an int, sensible names
2 Marks – constructor, 3 variables passed in, no type of method, variables assigned
2 Marks – encapsulation, getters and private variables
2 Marks – Sensible toString, returns a String
2 Marks – Equals method – not expecting much, 2 marks if a good attempt made,1.5 if didn't attempt to parse object first, 1 mark if didn't compare all 3 fields

```
public class Book {

    private String author;
    private int pages;
    private String title;

    public Book(String author, String title, int pages) {
        this.author = author;
        this.title = title;
        this.pages = pages;
    }

    public String getAuthor() {
        return author;
    }

    public int getPages() {
        return pages;
    }

    public String getTitle() {
        return title;
    }

    public String toString() {
        return author + ", Title: " + title +" , pages: " + pages;
    }

    public boolean equals(Book other){
        if(!this.author.equals(other.getAuthor())){
            return false;
        }
        if(this.pages !=other.getPages()){
            return false;
        }
        if(!this.title.equals(other.getTitle())){
            return false;
        }
        return true;
    }

}
```

b) Declare and initialise an array of length 5 with 5 books, populated with sensible values of your choice. You do not need to create the method headers. (2 marks)

```
Book[] books = new Book[5];

    books[0] = new Book("Asimov", "Foundation", 250);
    books[1] = new Book("Asimov", "Foundation and Empire", 100);
    books[2] = new Book("Asimov", "Second Foundation", 200);
    books[3] = new Book("Asimov", "Foundations Edge", 50);
    books[4] = new Book("Asimov", "Foundation and Earth", 210);
```

1 mark for array, 1 mark for sensible answers,

c) Create a method to identify and display the book with the highest number of pages, displaying the highest number of pages. You do not need to create the method headers. (4 marks)

```
int highest = books[0].getPages();

    for(int i = 0; i<books.length; i++){
      if(books[i].getPages()> highest )
        highest = books[i].getPages();
    }
    System.out.println("Highest: " + highest);
```

highest variable declared – 1 mark
For loop – 1 mark (0.5 if array. Length not used)(0.5 if highest not calculated)
calculate highest value – 1 mark
display highest pages – 1 mark

d) Create a method to identify and display the average number of pages for all the books in the array. You do not need to create the method headers. (4 marks)

```
double mean = 0;
    double total = 0;
    for(int i = 0; i<books.length; i++){
        total = total + books[i].getPages();
    }
    mean  = total/books.length;
    System.out.println("Mean page number: " + mean);
```

mean and total variables declared – 1 mark
For loop – 1 mark (for each is okay) (0.5 if array. Length not used)(0.5 if total not calculated)
calculate mean value – 1 mark

Question 3: (10 marks)

Complete the following Java private helper method to count the number of empty strings in an array of strings **using recursion**.

Do **not** use any loops. Submissions with loops will not pass.

Sample test cases:

```
String[] arr1 = {"123", "", "abc"};
System.out.println(countEmptyString(arr1));    → 1
String[] arr2 = {"", "", "", ""};
System.out.println(countEmptyString(arr2));    → 4
```

```
public static int countEmptyString(String[] arr) {
    return countEmptyString(arr, 0);
}
```

```
// private helper method
private static int countEmptyString(String[] arr, int start) {
    // base case


    // recursive step


  }
```

```
// private helper method
private static int countEmptyString(String[] arr, int start) {

    // base case
    if (start >= arr.length) {
        return 0;
    }

    // recursive step
    if (arr[start].length() == 0) {
        return 1 + countEmptyString(arr, start + 1);
    }
    return countEmptyString(arr, start + 1);

  }
```

Marking scheme
10 marks for passing all test cases
0 marks penalty for first incorrect submission
1 mark penalty for each subsequent incorrect submissions